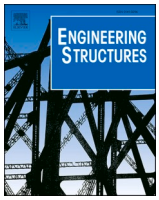
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Joint detection PCD-based method for automatic construction of geometric digital twin in cable dome structure

Ailin Zhang [a,b], Jie Wang [a], Yanxia Zhang [a,*], Guanghao Shangguan [a]

[a] Beijing University of Civil Engineering and Architecture, Beijing 100044, China
[b] Beijing University of Technology, Beijing 100124, China

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ABSTRACT

The geometric digitization of cable dome structures using Point Cloud Datasets (PCD) holds significant importance as it serves as a foundational step in constructing the Digital Twin of such structures. However, accurately detecting the positions of structural joints for effective construction of a geometric digital twin from PCD remains a challenging task. To address this issue, we propose an enhanced algorithm and system solution for joint detection based on target recognition. The method leverages a novel shape estimation model to identify structural components by randomly sampling the minimum set of points from the point cloud. Subsequently, connection joints and their coordinates are determined according to the spatial relationships of the components. This enables the digital reconstruction of the structure based on the topological relationships of the joints, ultimately facilitating the establishment of a geometric digital twin. The efficacy and reliability of the proposed method are validated through a case study involving PCD from a scaled cable dome structure, demonstrating outstanding performance in joint detection and fitting with accuracy levels exceeding 99 %. Based on the results of joint detection, the feasibility of model topology reconstruction is corroborated, and the potential for accurately assessing joint deviations is explored. These research findings serve to advance the progress of digital twins in large-span dome structures, facilitating enhanced analysis and inference regarding structural performance, safety, and stability within the digital twin framework.

1. Introduction

The cable dome structure, characterized by its efficient material usage, high load-bearing capacity, and ability to achieve large spans with elegance, finds extensive applications in numerous national landmark large-span public buildings. From the standpoint of engineering safety assessment, the substantial span of such structures renders joint deviations pivotal factors impacting prestress and structural stability [1,2]. The endeavor to construct a real-time analysis model of structural performance holds immense significance, with the crucial prerequisite being the establishment of accurate and efficient geometric Digital Twin (gDT) of the structures concerned. Digital twins (DTs) have demonstrated their significance in various phases of the building life cycle, including construction, maintenance, and reconstruction [3–7]. The foundation of gDT modeling rests upon creating a precise digital three-dimensional representation rooted in the physical realm. In the context of cable dome structures, the precise location of key joints stands out as the most crucial and tangible factor for mirroring the physical entity. The objective of developing a gDT serves a dual purpose: firstly, to enable rapid detection of joint deviations and early warning of potential security risks; secondly, the gDT serves as the fundamental model and primary initiative for mechanical DT projects. The construction of gDTs for tensioned cable dome structures rely on the PCD. This necessity arises from the presence of numerous joints within these structures, whose spatial coordinates cannot be swiftly measured by traditional methods, such as total stations. PCDs are characterized by its high precision, resolution, and dimensionality, offering rich geometric information that holds significant potential for extracting the intricate features of dome structures.

Current commercial point cloud processing solutions lack the inherent capability to generate gDTs for dome structures. Typically, these solutions focus on specific functionalities such as component shape recognition and segmentation. Moreover, these processes often involve semi-automatic or manual procedures (e.g., CloudCompare). Manual selection of similar points for joint position representation hampers automatic modeling, and establishing joint connections through

* Corresponding author.
E-mail address: zhangyanxia@bucea.edu.cn (Y. Zhang).

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geometric topology poses a significant challenge. Consequently, in the automation of joint detection for dome structures, the labor costs outweigh the potential benefits of DTs, significantly impeding the realization of true digital automation.

Establishing the gDTs of cable dome structures relies on accurately detecting components spatial relationship and fitting the spatial positions of key joints. Analogous to the detection of connectivity among various functional units in building structures, topological information effectively facilitates the construction of the building's gDT [4]. Recent research indicates that detecting and fitting joints in large-span steel structures yield various practical applications. For instance, determining the degree of structural components bending based on joint coordinates [8], utilizing fitting results for joint segmentation and topological information in establishing BIM models for dome structures [9], and deriving construction errors from joint position coordinates [10]. Consequently, the spatial location and topological relationships of joints constitute a pivotal step in the digitization of cable dome structures.

The process of structural digitization, particularly in the context of cable dome structures, necessitates the detection and segmentation of target objects, along with the establishment of digital spatial relationships. Target detection and segmentation involve identifying the geometric entity shape within the PCDs and segmenting the object based on the inter-entity relationships. In the case of dome structures, the target object encompasses all the constituent components forming the structure. While existing methods can effectively detect and classify certain types of geometric objects, they currently lack the capability to automatically fit key joints based on the detection and segmentation results, or to automatically establish the connection relationships between joints.

Therefore, we propose the introduction of the concept of a gDT for cable dome structures. Central to this is the pivotal challenge of joint position detection, which essentially entails the precise fitting of spatial coordinates and positional relationships of all joints based on PCD. To address this challenge, we present a comprehensive process that spans from the extraction of low-level geometric information to the establishment of high-level spatial structural topology, as depicted in Fig. 1. The extraction of geometric information encompasses target object detection, component segmentation, joint coordinate fitting. Meanwhile, spatial structural topology relies on the spatial topological relationships among joints to establish gDTs. Notably, our proposed method integrates joint fitting and inter-joint relationship connection within the same solution, thereby significantly advancing the realization of digital automation. In summary, our contributions are outlined as follows:

- A proposal is outlined for constructing a gDT for cable dome structures, based on joint position detection. This solution integrates target detection, segmentation, joint fitting, and automatic topology reconstruction functionalities.
- The minimum set shape estimation model of point cloud is proposed innovatively. This model ensures computational accuracy and efficiency while mitigating the impact of noise points.
- Innovatively introduced into the target recognition framework are the random sampling constraint mechanism and adaptive termination mechanism, enhancing efficiency and precision.
- We validate the effectiveness of the proposed method through a case study and explore its potential applications in joint deviations.

2. Background and related work

Digital twin technology offers an effective and dependable approach for the rapid and precise detection of structural joint positions. It facilitates the transformation of safety detection from labor-intensive processes to digital automation, while also supporting the information management of large-span structures. However, challenges remain in creating DTs for cable dome structures and meeting the demands of joint detection. Traditional methods like total station measurements have not met the desired efficiency levels [11]. Joint detection using PCD is mainly manual or semi-automated [12]. There is an urgent requirement to develop algorithms that can meet the needs of digital processing and information management in structural domains.

For the joint detection in cable dome structures, the fundamental objective lies in detecting the target object accurately. Currently, methods for target object detection can be categorized into model-driven and data-driven approaches. The model-driven method achieves object recognition and segmentation in PCD by optimizing parameter input according to the mathematical representation of the target object. For instance, a predominant feature of cable dome structures comprises tubular components, effectively represented as cylinders within PCD. Automated techniques, including RANSAC [13,14], Hough transform [15], and region growth segmentation [16], are employed for object segmentation by leveraging original shape detection methodologies. The connectivity or adjacency of distinct target objects is established through joints, enabling the realization of joint fitting and coordinate extraction subsequent to object segmentation. Data-driven detection involves inferring patterns from extensive sets of labeled data to detect objects. Research indicates that machine learning can enable the digital processing of building structures and direct object detection from large-scale point clouds [17–19]. While this approach is not confined by specific environmental constraints, enhancing the recognition success rates of diverse objects necessitates the expansion of the training dataset.

The application of the conventional model-driven approach to point cloud target detection and segmentation has demonstrated favorable

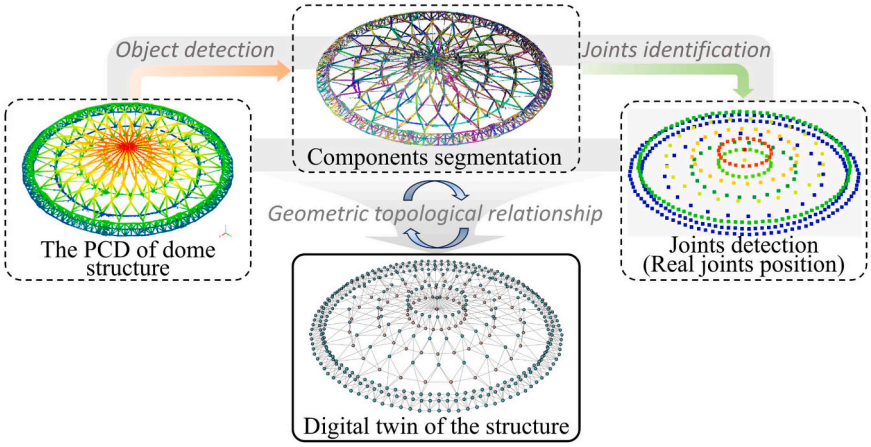


Fig. 1. Geometric digital twin construction of cable dome based on PCD.

performance [8,9,20]. These investigations involve the explicit simulation and segmentation of target objects, along with the establishment of inter-object connections. Moreover, the model-driven methodology facilitates joint coordinate fitting in subsequent stages and encapsulates essential information for deriving structural topology. Nevertheless, existing research predominantly centers on reverse modeling or deformation identification [21–23], neglecting structural joint detection and thereby impeding the creation of a true gDT [24]. Simultaneously, the mentioned methods cannot be directly applied to establish the gDT of the cable dome structure. The RANSAC algorithm stands out as one of the most widely employed robust estimation tools. Its widespread adoption stems from not only its simplicity but also its capacity to furnish reliable parameter estimation even in scenarios where more than half of the data comprises outliers. Nevertheless, while the basic RANSAC algorithm is robust, further enhancements are imperative to elevate its accuracy, efficiency, and stability [25,26].

The process of topology reconstruction, predicated on joint detection outcomes, represents an ongoing endeavor aimed at automatically interlinking an explicit three-dimensional geometric model through the topological associations within joint space to facilitate visualization. Effectively addressing this challenge hinges on establishing joint correspondences and articulating them through mathematical frameworks. Prior research has employed direct array or sequential representations to express the planar distribution of joints, proving highly efficient and rapid in delineating an optimal structural geometric model [27]. Nevertheless, the geometric model constructed based on the detected joint coordinates cannot be directly resolved using the above method due to the inherent deviation of joints from their intended positions. Additionally, topology reconstruction can be applied by integrating model information directly into finite element software for efficient 3D geometric modeling, reducing manual effort and time needed for modeling tasks.

According to related work, the following research gaps have been summarized:

- Currently, there is no comprehensive and systematic fully automated method for detecting joint positions and establishing geometric digital twins of cable dome structures.
- Data-driven object detection struggles to achieve ideal performance with sparse datasets. Model-driven approaches, such as traditional RANSAC algorithms, fail to meet the accuracy and efficiency requirements for detecting joints in cable dome structures.
- Implementing automated 3D topology reconstruction based on detected joint coordinates is also challenging. Deviations in actual joint positions prevent direct application of theoretical topology algorithms for geometric modeling.

This paper aims to provide a systematic and universally applicable method for the geometric digital twin (gDT) of cable dome structures, while addressing the aforementioned gaps.

3. Proposed solution

A geometric digital twin solution (abbr. DS-gDT) for constructing cable dome structures is proposed. It expects a PCD as input and, through a series of developed algorithms, output joint spatial position

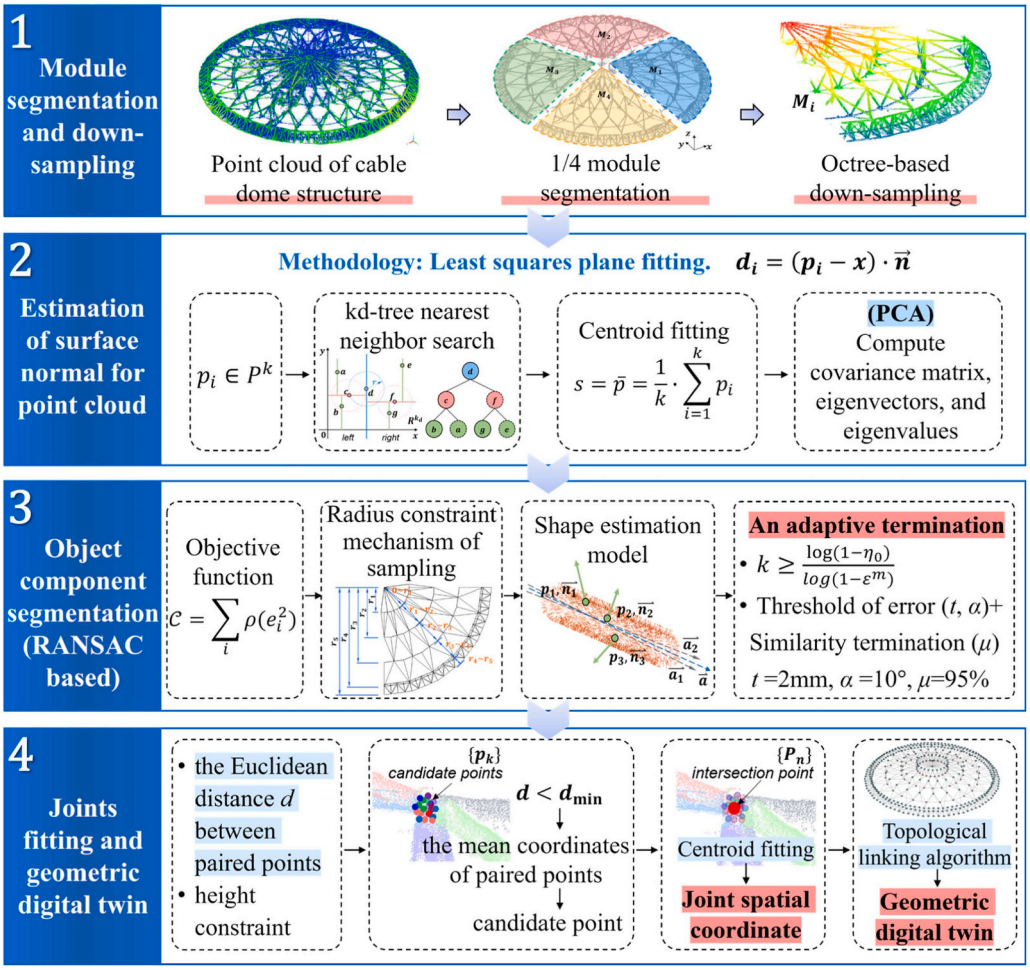


Fig. 2. Solution for detecting joints PCD-based.

coordinates arranged in a specific order and the gDT of structure. Fig. 2 illustrates our proposed method, encompassing the following sequential steps: (1) Module segmentation and down-sampling, (2) Estimation of surface normal for point cloud, (3) Object component segmentation, and (4) Joints fitting and geometric digital twin. The detailed elaboration of each step will be presented in subsequent sections.

3.1. Module segmentation and down-sampling

To enhance the precision and fault tolerance of the final joint coordinates, a method is proposed to partition the PCD into 1/4 modules, as illustrated in Eq.3 and Eq. (1). Adjacent modules are allowed to have an angular adjustment of 2° to 3°, and this angle is determined through multiple experiments. The principle underlying this determination is to avoid duplicate fitting of identical joints.

$$M = \{M_1, M_2, M_3, M_4\} \tag{1}$$

M represents the entire PCD of structure, while $\{M_i\}$ (i =1,2,3,4) represents each 1/4 module.

Considering the vast scale of initial point cloud, down-sampling is necessary to compress data and reduce computational costs. Currently, there are several classical point cloud down-sampling methods such as voxel grid down-sampling, uniform down-sampling, and octree down-sampling. Voxel grid down-sampling computes the centroids of point clouds within each grid cell. Consequently, after down-sampling, the positions of points in the point cloud may change, as they represent centroids or average positions of points within respective grid cells, rather than specific points identified by their original IDs in the initial point cloud (as shown in Fig. 4a). While uniform down-sampling theoretically preserves original positions, it may exhibit non-uniform effects across different regions, as demonstrated in the experimental results of this study (Fig. 4b) demonstrate this phenomenon. Octree down-sampling recursively subdivides the point cloud into parent and child relationships, demonstrating superior down-sampling effectiveness compared to the above methods. Initially, the maximum scene size is determined, and the first cube is established. Unit elements are sequentially placed into cubes that can contain them without child nodes. If a cube exceeds the minimum voxel size, it is subdivided into eight equal parts, distributing all unit elements within it among its eight child cubes. Subdivision stops for a child cube containing a nonzero number of unit elements identical to its parent cube. All data points associated with octree cells are replaced by the cell's center, effectively compressing these elements into a single octree cell. Fig. 5 illustrates the application of octree-based down-sampling. It should be noted that if there are a significant number of noise points in the PCD, noise point reduction can be performed using algorithms such as radius filtering, statistical filtering, and Gaussian filtering before down-sampling.

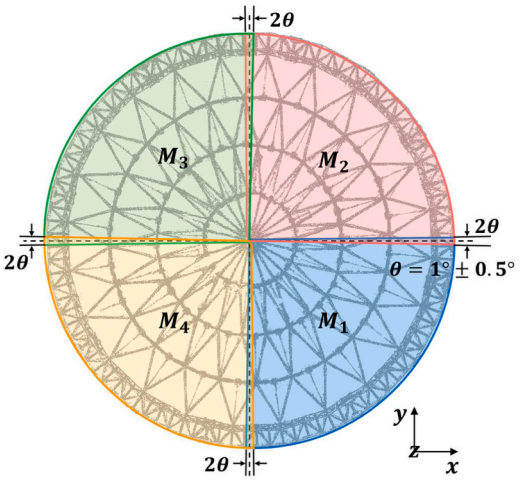


Fig. 3. The principle of module segmentation.

3.2. Estimation of surface normal for point cloud

The normal vector serves as a crucial attribute for each point within PCD, playing a vital role in the accurate estimation of normal vectors required for the point cloud segmentation algorithm essential in target object recognition. The determination of surface normal vectors within the point cloud necessitates neighboring point support, with the neighborhood's size typically denoted by either the neighborhood radius or the count of adjacent points. In this paper, resolving the normal vector of a surface point in the point cloud parallels the task of estimating the normal to a tangent plane on the surface, thereby transforming into a least squares plane fitting estimation problem. The resolution of the surface normal estimation problem ultimately boils down to computing the eigenvalues and eigenvectors of the covariance matrix through PCA (Principal Component Analysis) dimensionality reduction.

Given a point ($p_i \in P^k$), the utilization of a kd-tree nearest neighbor [28] search employing the depth-first search (DFS) algorithm facilitates the acquisition of neighboring points of p_i. Subsequently, the centroid s is computed as depicted in Eq. (2). Consequently, the plane P_l fitting procedure, derived from these neighboring points and employing the least squares method, can be expressed by Eq. (3):

$$s = \bar{p} = \frac{1}{k}\sum_{i=1}^{k} p_i \tag{2}$$

$$P_l = \min_{s,\vec{n},\|\vec{n}\|=1} \sum_{i=1}^{k} \left[(p_i - s)^T \vec{n}\right]^2 \tag{3}$$

Where, $\vec{n}$ is the normal vector of the fitting plane.

Within the optimization objective function, the goal is to identify a plane encompassing all neighboring points such that these points exhibit minimal distances from it. Therefore, the selection process involves isolating the minimum eigenvalues along with their corresponding eigenvectors from the covariance matrix. To elaborate further, for each point p_i, the associated point covariance matrix C is defined as follows:

$$C = \frac{1}{k}\sum_{i=1}^{k} \varepsilon \cdot (p_i - \bar{p}) \cdot (p_i - \bar{p})^T \tag{4}$$

$$C \cdot \vec{v}_j = \lambda_j \cdot \vec{v}_j, j \in \{0, 1, 2\} \tag{5}$$

Among them, λ_j represents the *j-th* eigenvalue of the covariance matrix, and $\vec{v}_j$ signifies the *j-th* eigenvector of the covariance matrix. Consequently, the normal vector of the fitting plane $\vec{n}$ is determined by the eigenvector corresponding to the minimum eigenvalue.

3.3. Object component segmentation

The RANSAC (Random Sample Consensus) paradigm extracts shape primitives by randomly selecting the minimum set of points from the point cloud data and constructing the corresponding shape primitives. The minimum set refers to the minimum number of points required to uniquely define a geometric primitive of a given type. Model equation parameters and error rates are computed using randomly selected points to determine whether the established model is optimal, and iterative calculations are performed until the best model is achieved. The geometric feature model of interest in this study is the cylinder. The traditional cylinder fitting method involves using two points with normal lines and judging the number of inner points based on the distance threshold to obtain the best fitted cylinder [13]. However, this method is greatly influenced by the quality of the point cloud. In cases where sampling times are consistent, the data set of the cylindrical model

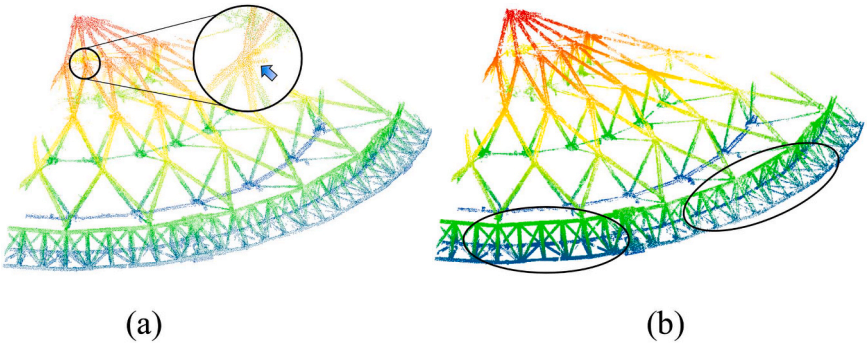


Fig. 4. (a) Results of voxel grid down-sampling. (b) Results of uniform down-sampling.

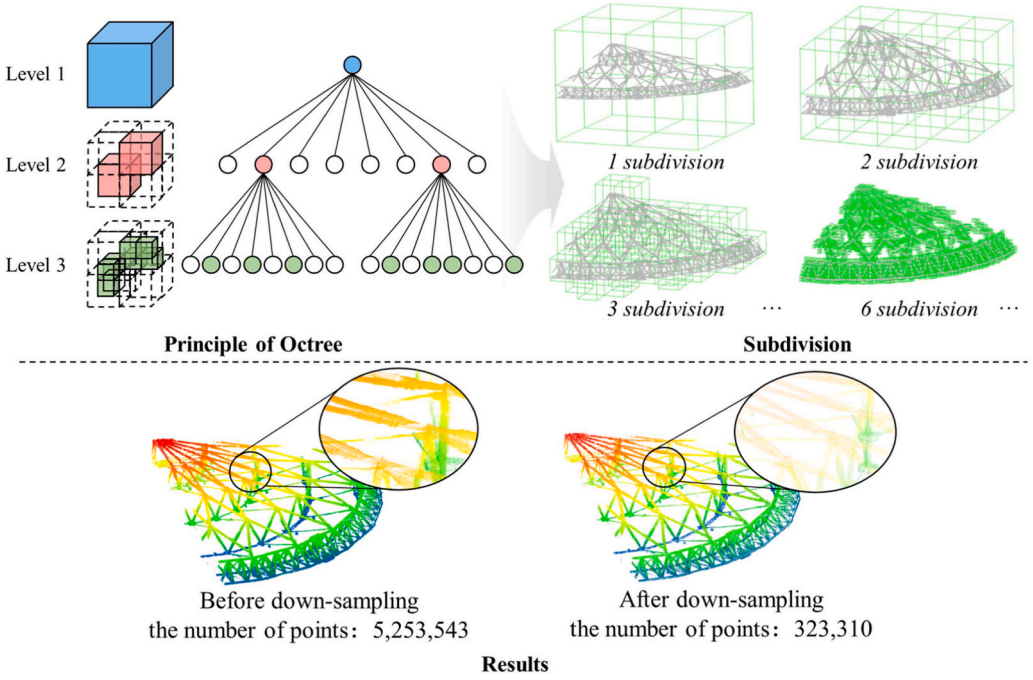


Fig. 5. Octree-based down-sampling.

satisfying the judgment criterion is reduced, reducing the probability of obtaining the optimal model. Therefore, this paper proposes a "3-point normal" cylinder fitting model (abbr. T-Ncf), whereby three points are randomly selected, neighborhood points are obtained using kd-tree nearest neighbor search, and PCA dimensionality reduction is utilized to solve the covariance matrix and obtain the normal vector of the points. The parameters are calculated using the proposed three-point normal cylinder fitting model. Distance and angle thresholds are used as criteria to judge the optimal model parameters, resulting in accurate identification and segmentation of the target object.

3.3.1. *"3-point normal" cylinder fitting model*

Specifically, as shown in Fig. 6, the normal vectors $\vec{m}_1$, $\vec{m}_2$, $\vec{m}_3$ of the randomly selected three points p_1, p_2, p_3 are computed using the method described in Section 3.2. These normal vectors are then utilized to construct the initial values of $\vec{v}_1$, $\vec{v}_2$ in the axis direction, as shown in Eq. (6).

$$\begin{cases} \vec{v}_1 = \vec{\tau}_1 \times \vec{\tau}_2 \\ \vec{v}_2 = \vec{\tau}_2 \times \vec{\tau}_3 \\ \phi = \arccos(\vec{v}_1 \times \vec{v}_2) \end{cases} \tag{6}$$

ϕ represents the construction axis angle. In theory, $\vec{v}_1 = \delta \vec{v}_2$, where

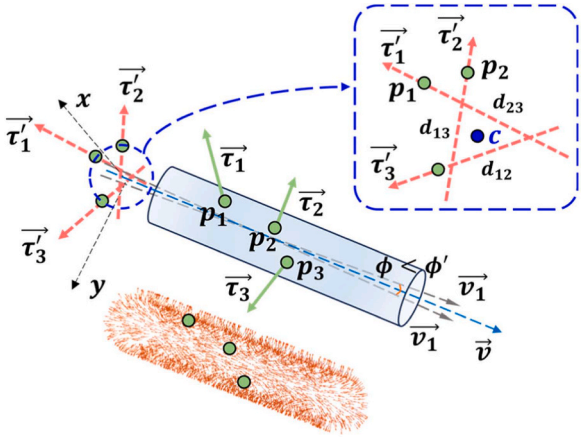


Fig. 6. Diagram of the "3-point normal" cylinder fitting model.

δ is a constant. However, due to the presence of noise on the cylinder surface in point cloud data, an angle exists between the two axis directions. If $\phi < \phi'$ (where ϕ' is the angle threshold), then the central axis

of the entire cylinder can be represented as $\vec{v} = \vec{v}_1 + \vec{v}_2$.

The three parameter lines, $p_1 + m\vec{\tau}_1$, $p_2 + m\vec{\tau}_2$, and $p_3 + m\vec{\tau}_3$ are projected onto the $\vec{v}x = 0$ plane along the axis $\vec{v}$. In the presence of noise in the point cloud data, these projections intersect pairwise, forming a triangle. Illustrated in Fig. 6, the lengths of the triangle's edges are denoted as d_{12}, d_{13}, and d_{23}. If Eq. (7) is satisfied, the centroid of the triangle, c (computed using Eq. 8), is determined. Otherwise, the data set is reselected using the aforementioned steps for model parameter calculation.

$$\{d_{12}, d_{13}, d_{23}\} < d_0 \tag{7}$$

$$\begin{cases} x_c = \frac{1}{3}\sum_{i=1}^{3} x_i \\ y_c = \frac{1}{3}\sum_{i=1}^{3} y_i \\ z_c = \frac{1}{3}\sum_{i=1}^{3} z_i \end{cases} \tag{8}$$

Here, d_0 is the distance error threshold. The coordinates of the centroid c are denoted as (x_c, y_c, z_c), while the three intersection points of the parameter lines' projections are represented as (x_i, y_i, z_i).

To enhance the efficiency of the RANSAC algorithm in the random sample selection process, a *radius constraint mechanism* is incorporated to guide the random sampling process, considering the planar geometric characteristics of the dome structure. As illustrated in Fig. 7, the plane's radius is segmented into $r_1 \sim r_5$, and the sampling area is delineated based on the radius intervals: $\{0 \sim r_1\}$, $\{r_1 \sim r_2\}$, $\{r_2 \sim r_3\}$, $\{r_3 \sim r_4\}$, $\{r_4 \sim r_5\}$. This approach ensures that the minimum set and model parameters are randomly selected within the defined sampling areas. The radius constraint mechanism helps maintain the quality of random samples and mitigates the occurrence of "over-acquisition", which involves capturing sample points with abrupt changes in curvature, while also reducing computational costs to a certain extent.

3.3.2. An adaptive termination

The termination criterion of the traditional RANSAC algorithm incorporates an adaptive termination (AT) criterion by updating the number of iterations N using the iterative interior point rate ε. When N exceeds a predetermined threshold, the process is terminated (Eq. 9). The interior point ratio refers to the ratio of the number of interior points T to the total number of points M. The subset with the highest number of interior points is considered the best matching point, which leads to the generation of the best model. Since N depends on ε, the interior point rate directly influences the termination process. When the number of interior points is small, it increases the likelihood of selecting a local optimum. To improve efficiency and result accuracy, a *double termination criterion* is adopted. This criterion combines the traditional adaptive termination criterion with a similarity criterion and distance threshold. The similarity criterion considers the similarity of interior points between two consecutive iterations. If the similarity exceeds 95 %, the algorithm terminates; otherwise, it continues until a certain threshold is met.

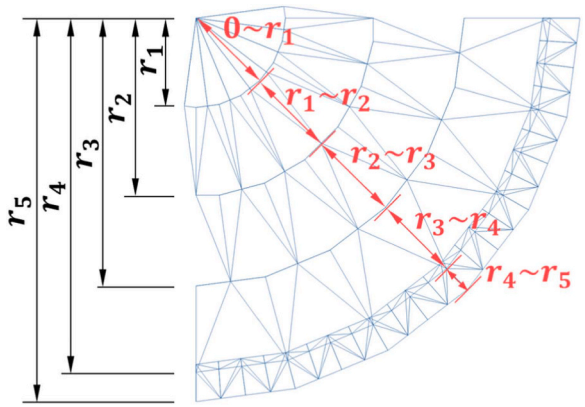


Fig. 7. Radius constraint mechanism for random sampling.

$$N \geq \frac{\log(1-\eta_0)}{\log(1-\varepsilon^e)} \tag{9}$$

$$\varepsilon = \frac{T}{M} \tag{10}$$

Where, e represents the minimum sample size of each subset, indicating the minimum number of points required to calculate the model parameters. In this paper, the value of e is set to 3.

The distance threshold plays a crucial role in determining the support of a specific model within a dataset. It distinguishes whether a data point is considered an inlier or an outlier based on its distance from the model. While empirical approaches can be used to determine such thresholds in practice, a more formalized method can be employed. Following the methodology outlined in [29], the distance error can be assumed to follow a Gaussian distribution with a mean of 0 and a standard deviation of σ. In this case, the squared perpendicular distance ($d_\perp^2$) of a point is the sum of squares of Gaussian variables (Eq. 11). This squared distance follows a χ_s^2 distribution with s degrees of freedom, where s corresponds to the codimension of the model. The probability that a χ_s^2 random variable, denoted as ξ, is less than t^2 is determined by the cumulative chi-square distribution (Eq. 12). By utilizing the inverse chi-square distribution, a threshold value, denoted as t, can be determined to capture a specific proportion of true inlier points, as illustrated in Eq. (13).

$$d_\perp^2 = (x_i - x_j)^2 + \left(y_i - y_j\right)^2 + (z_i - z_j)^2 \tag{11}$$

$$F_s(t^2) = \int_0^{t^2} \chi_s^2(\xi) d\xi \tag{12}$$

$$\begin{cases} \text{inlier} d_\perp^2 < t^2 \\ \text{outlier} d_\perp^2 \geq t^2 \end{cases} \quad \text{with } t^2 = F_s^{-1}(\alpha)\sigma^2 \tag{13}$$

Where, (x_i, y_i, z_i), (x_j, y_j, z_j) denote the coordinates of two points, α typically takes a value of 0.95, indicating that an interior point will only be erroneously rejected in 5 % of cases.

3.4. Joints fitting and geometric digital twin

3.4.1. Joints fitting

Cable dome structures typical include upper joints, cable bracing joints, upper chord joints of the ring truss, and lower chord joints of the ring truss. The identification and segmentation of these components serve as a fundamental step in elucidating the spatial positioning relationships of each constituent part. Furthermore, it is an essential prerequisite for extracting the three-dimensional coordinates of the intersection points, referred to as joints. This section presents the proposed process for extracting three-dimensional coordinates and the key algorithms pertaining to joints, based on the outcomes of component identification and segmentation, as illustrated in Fig. 8.

The specific details of the algorithm flow are explained as follows:

(1) The algorithm initially selects two data sets, S_i and S_j, from the entire set of segmented component point clouds. To mitigate noise interference, the algorithm verifies whether the number of data sets exceeds 200. If this condition is not met, the selection process is repeated.

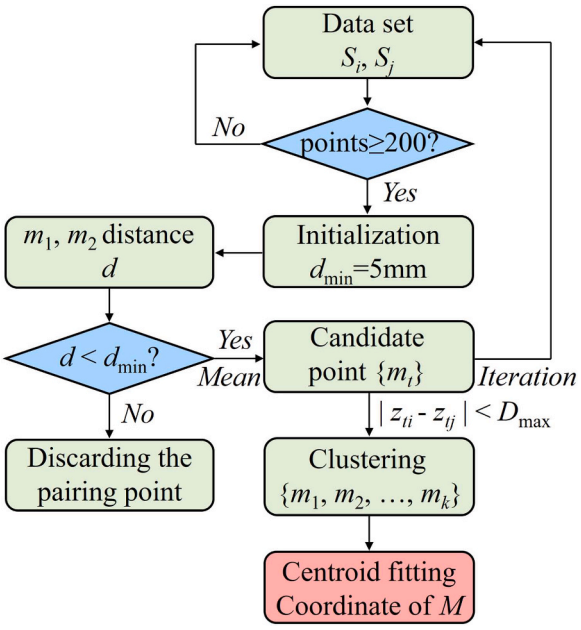


Fig. 8. Three-dimensional coordinate extraction process of joints.

(2) Initialization of the minimum distance variable, d_{min}, is set to 5 mm.
(3) Points m_1 and m_2 are selected from S_i and S_j, respectively, referred to as "pairing point", and the Euclidean distance d between the pairing points is calculated using Eq. (14). If $d < d_{min}$, the mean coordinate is computed according to Eq. (15) to yield the "candidate point $\{m_t\}$". Alternatively, if $d \geq d_{min}$, the group pairing points are discarded. This process is repeated to obtain all candidate points.
(4) The candidate points are clustered using Eq. (16), such that k candidate points $\{m_1, m_2, \ldots, m_k\}$ within each joint range are allocated to the same attribute group based on explicit distance constraint criteria.
(5) The centroid of the candidate points within each cluster group is computed using Eq. (17) to derive the three-dimensional coordinates of the joint m (refer to Fig. 9).

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \tag{14}$$

$$\{m_t\} = \begin{Bmatrix} (x_1 + x_2)/2, \\ (y_1 + y_2)/2, \\ (z_1 + z_2)/2 \end{Bmatrix} \tag{15}$$

$$\left| z_{ti} - z_{tj} \right| < D_{max} \tag{16}$$

$$\begin{cases} x_m = \dfrac{1}{k}\sum_{n=1}^{k} x_{tn} \\ y_m = \dfrac{1}{k}\sum_{n=1}^{k} y_{tn} \\ z_m = \dfrac{1}{k}\sum_{n=1}^{k} z_{tn} \end{cases}, \; M = \begin{pmatrix} \\ x_m, y_m, z_m \\ \\ \end{pmatrix} \tag{17}$$

Among them, the coordinates of the matching points m_1 and m_2 are denoted as (x_1, y_1, z_1) and (x_2, y_2, z_2), respectively. The candidate point coordinates are represented as (x_t, y_t, z_t), where z_{ti} and z_{tj} denote the z coordinates of different candidate points. Additionally, D_{max} corresponds to the maximum diameter of the component at the same intersection point.

Due to the significance of selecting the "pairing point" located in the upper or lower part when the intersection of two components occurs, it becomes meaningless to choose a point situated in the middle of the component. To enhance the computational efficiency of the algorithm, a stringent constraint condition is introduced into the "pairing point" selection process. The "pairing region" is determined based on the z-direction (i.e., the height direction) of the point, as depicted in Fig. 10. There are two cases to consider: (a) in the case of a vertical component, the height range of the pairing area is $[(z_1\text{-}t), z_1]$; (b) for non-vertical components, the height range of the matching area is $[(z_1\text{-}3t), (z_1\text{-}2t)]$, where z_1 represents the maximum z coordinate of the component, z_2 denotes the minimum z coordinate of the component, and t is the constraint parameter. Through extensive testing, the value of t is computed in accordance with Eq. (18).

$$t = \frac{1}{6}\left(z_1 - z_2\right) \tag{18}$$

3.4.2. Geometric digital twin

The primary distinguishing characteristic among diverse cable dome structures resides in the quantity of joints they encompass. Thus, the algorithm functions by regarding the joint count as the input variable, applying specific criteria to assign numerical labels to the joints, leveraging the adjacency matrix for constructing the joint network model, and interconnecting the relevant joints in accordance with geometric topological relationships. This systematic approach results in the formulation of an integrated automatic joint topology connection algorithm tailored for dome structures, thereby enabling the creation of a gDT.

Joint numbering follows a prescribed set of rules, specifically adhering to the principle of "from the inside out, from high to low", as illustrated in Fig. 11. The concept of "from inside to outside" dictates that joint numbering progresses from the innermost circle to the outermost circle based on the radial distribution along the joint circumference. For instance, in the blue section of the Fig. 11, the numbering sequence begins with the first circle denoted as $\{1, 2, \ldots, a_1\}$, followed by the second circle designated as $\{a_1+1, a_1+2, \ldots, \sum_{i=1}^{2} a_i\}$, and so forth. Subsequently, the nth circle is assigned numbers $\{\sum_{i=1}^{n-1} a_i +$

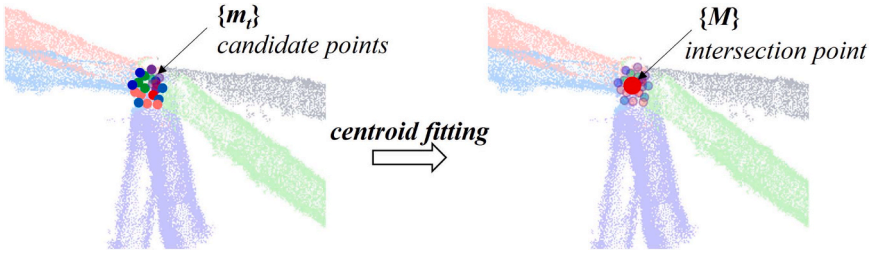


Fig. 9. Coordinate fitting of joint.

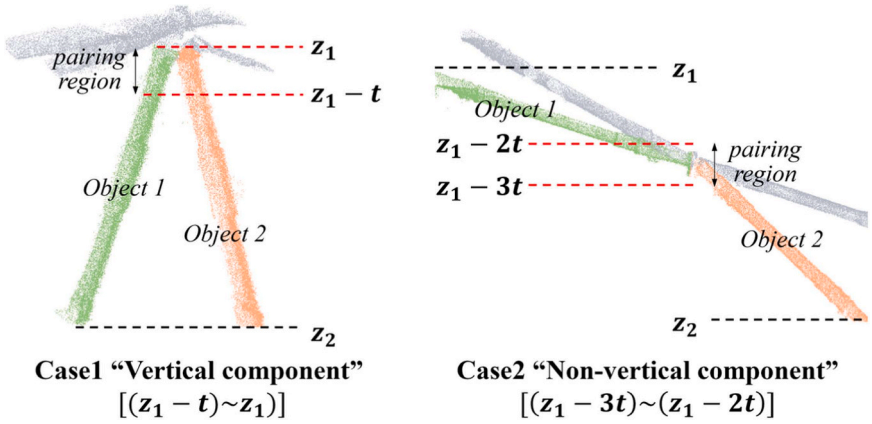


Fig. 10. The height constraint of the pairing region.

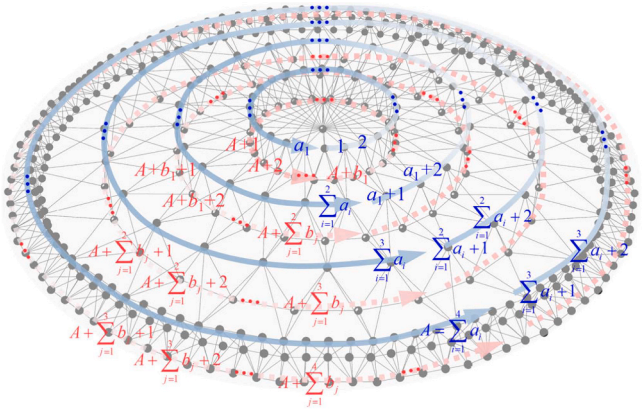


Fig. 11. Numbering principle for joints.

$1, \sum_{i=1}^{n-1} a_i + 2, \ldots, \sum_{i=1}^{n} a_i\}$, where a_i represents the joint count in each circle, n denotes the total number of radial joint circumferences, and i ranges from 1 to n. "From high to low" signifies following the numbering sequence from the innermost to the outermost joints with greater vertical elevation along the radial circumference, the enumeration then progresses to include joints with lesser vertical elevation. For instance, the blue segment in the Fig. 11 represents joints with greater vertical elevation, while the red segment denotes joints with lower vertical positioning. Here, A denotes the total count of joints with greater vertical elevation, and b_i symbolizes the quantity of joints within each circle possessing lower vertical positioning.

The emphasis lies in the binary connectivity of joint pairs, without consideration for joint differentials or the intensity of their relationships. As such, the entire cable dome structure is conceptualized as an unweighted, undirected network topology (denoted as D) comprising joints and edges. Each intersection of the structural components corresponds to a joint, while each component represents an edge, articulated in set notation as Eq. (19):

$$D = \{J, L\} \tag{19}$$

Here, the nonempty set $J = \{j_1, j_2, \ldots, j_n\}$ pertains to the collection of joints within D, while the set $L = \{l_1, l_2, \ldots, l_n\}$ encapsulates the edges present in D. Each edge $l_i = \{j_s, j_k\}$ within this context correlates to a subset comprising two elements from the set J.

To streamline program implementation, an adjacency matrix is employed to delineate the unweighted, undirected network topology, with the matrix dimension aligning with the joint count. Given that the undirected topology graph D encompasses n joints, an $n \times n$ adjacency matrix A (Eq. 20) is utilized to depict inter-joint connectivity. The element value a_{ij} within the adjacency matrix A, as depicted in Eq. (21), signifies the connection status between joints i and j.

$$A = [a_{ij}] \tag{20}$$

$$a_{ij} = \begin{cases} 1, & \text{Joints} j_s \text{and} j_k \text{areconnected.} \\ 0, & \text{Joints} j_s \text{and} j_k \text{arenotconnected.} \end{cases} \tag{21}$$

4. Experiment and results

In this section, we will assess the practical application and efficacy of the proposed methodology via a case study. The chosen case is a scaled model of a fully assembled cable dome structure, with pertinent structural design details available in the references [30,31]. The holographic point cloud data of the structure was acquired using a laser scanner (Fig. 12).

4.1. Point cloud processing

Utilizing the module segmentation principle and down-sampling method, the acquired PCD of the structure undergoes processing. Initially, the PCD is segmented into four modules (M1∼M4), as depicted in Fig. 13(a). To streamline the subsequent gDT establishment, all joints are systematically numbered following the methodology outlined in Section 3.4.2 prior to module subdivision. Subsequently, each module undergoes down-sampling using the Octree method. The down-sampling results are depicted in Fig. 13(b), showcasing the quantity of point clouds before and after the down-sampling process for each module. Our approach leverages the Octree framework (OctFilter / addPointsFromInputCloud) within PCL, assigning a unique identifier to each voxel. Following extensive experimentation, a voxel size of 2 mm × 2 mm × 2 mm is selected, and the Octree resolution is fixed at 128 for optimal results. An inherent advantage of this approach lies in the hierarchical organization of PCD facilitated by the Octree data structure, enhancing down-sampling efficiency while mitigating shape distortion or information loss arising from random sampling. The results of module segmentation reveal the absence of duplicate joints across modules, indicating that module segmentation occurs solely at the component level rather than among adjacent joints within each module. With geometric shape features remaining unaltered, down-sampling results in approximately a 98 % reduction in the number of points within each module's point cloud. This reduction significantly optimizes memory utilization and enhances algorithmic efficiency in subsequent stages.

4.2. Components identification and segmentation

The estimation of point cloud surface normal is computed internally within the program. Further definition of the direction of the normal

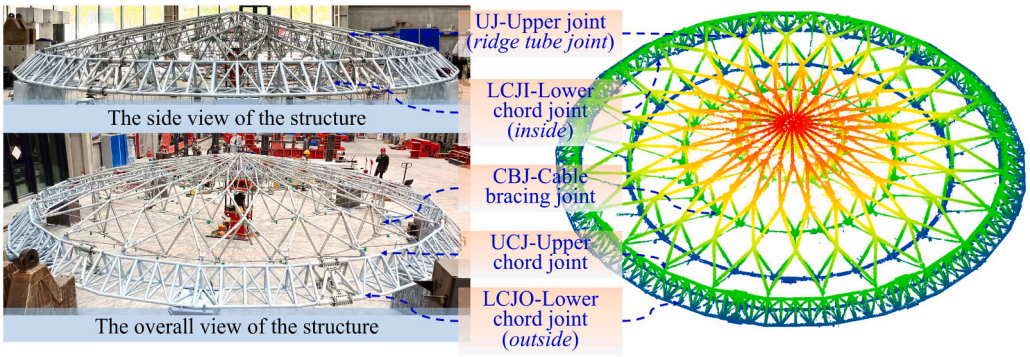


Fig. 12. Case: scale model of cable dome structure and its point cloud.

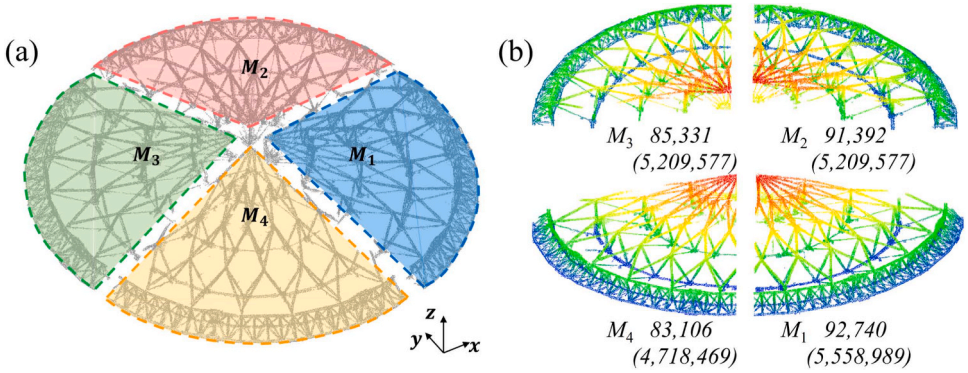


Fig. 13. (a) PCD module segmentation, (b) Result of the down-sampling.

vectors is required, namely, the normal vector $\vec{n}$ at point p aligns with the viewing direction V (Eq. 22).

$$\vec{n}\cdot(V_p - p) > 0 \tag{22}$$

Using Module 1 as a case study, the point cloud of Module 1 is processed through DS-gDT, employing the shape fitting model T-Ncf, radius constraint mechanism sampling, and a double termination criterion (with a selected distance threshold of 2 mm, and an angle threshold of 10°). The results are visualized in Fig. 14(a), where distinct colors represent different components. Analysis of the component recognition rate by DS-gDT revealed a recognition success rate exceeding 99 % for all components. Despite challenges such as down-sampling, interference occlusion, and laser diffusion impacting the recognition of certain components, a minimal number of identification failures were tolerated, as they do not affect the final joints fitting outcome.

4.3. Construction of geometric digital twin

Initially, the joint fitting of the segmented point cloud is executed using the three-dimensional coordinate extraction algorithm for joints. As illustrated in Fig. 15(a), the calculation results of candidate points (using Module 1 point cloud as an example) demonstrate that all candidate points form a clustered group within the joint range based on clustering conditions. The centroid of each cluster group of candidate points is computed to obtain the three-dimensional spatial coordinates of the joints, as depicted in Fig. 15(b). It is evident from the outcomes that the proposed algorithm adeptly fits all joints and their coordinates. Despite a lower cable recognition rate, the algorithm exhibits commendable stability and reliability.

The fitting joints are organized based on their assigned numbers, forming the joint set $[M]=[M_{ij}]$. Utilizing the topological characteristics of these joints, an equivalent network model mirroring the actual structure is constructed, depicted in Fig. 16(a). This network model serves to elucidate the topological interconnection among joints within

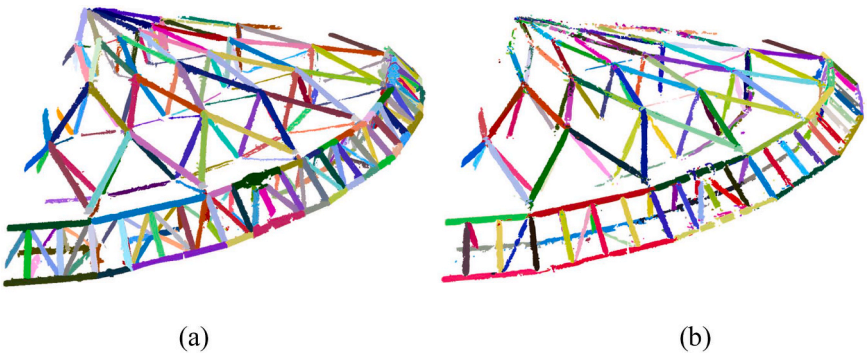


Fig. 14. (a) Segmentation result based on DS-gDT. (b) Segmentation result based on traditional RANSAC.

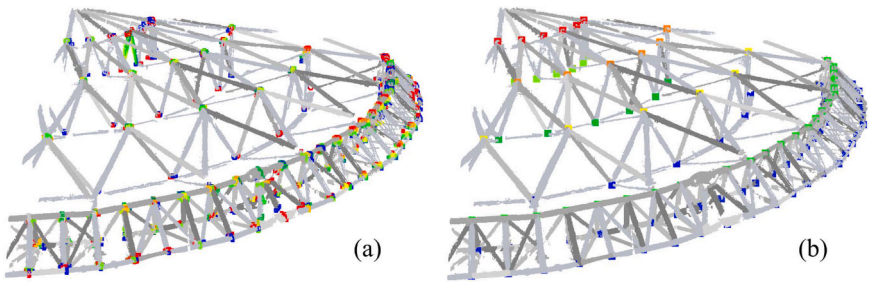


Fig. 15. (a) Candidate points for Module 1. (b) Joints fitting of Module 1.

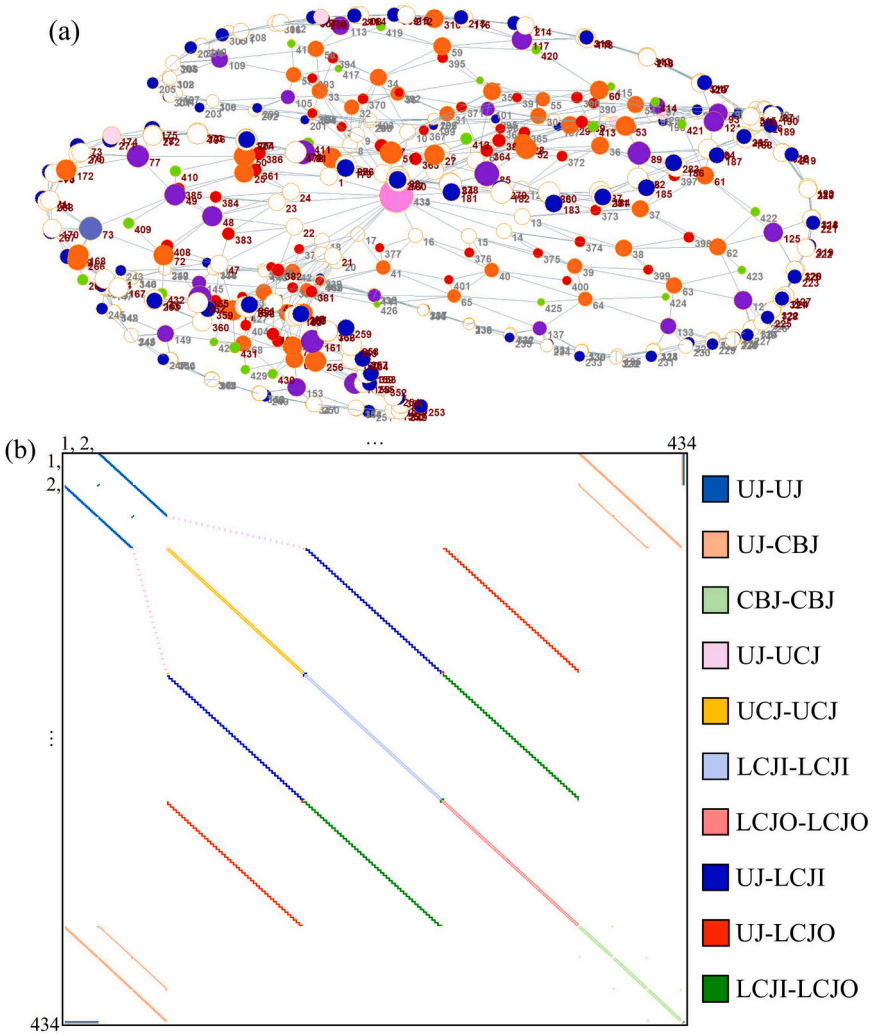


Fig. 16. (a) Equivalent network model. (b) The corresponding visual adjacency matrix, and the joint codes can be referenced in Fig. 12; for example, "UJ-UJ" denotes "upper joint is connected to upper joint".

the cable dome structure. The size of a joint's diameter in the model directly correlates to the number of components linked to that joint. An adjacency matrix is employed to represent this network model, illustrated in Fig. 16(b), where a "black dot" signifies a connection between two joints, while a blank indicates no connection. Within the program, joint connectivity is determined through relevant logic and algorithms, involving indexing and accessing elements in the adjacency matrix. Explicitly solving the adjacency matrix is often unnecessary.

Given that the cable dome structure in this case comprises two distinct sections: the upper structure (cable dome) and the lower structure (ring truss), the process of establishing the gDT for both upper and lower structures is delineated following the methodology outlined in Section 3.4.2 (as depicted in Fig. 17). Despite the structural division into two parts for developing gDTs, an inherent relationship exists between them: the outermost joints of the cable dome align with the upper chord joint of the ring truss at the corresponding position. Thus, the fusion of the two sections in the final gDT necessitates no additional assessment of the interconnection status of coinciding joints. Furthermore, the adjacency matrices of the two sections adhere to the relationship defined by Eq. (23), wherein the sum of the rank of the cable dome joints adjacency matrix A_d and the ring truss joints adjacency matrix A_t equals the rank of the overall joint set $[M]$.

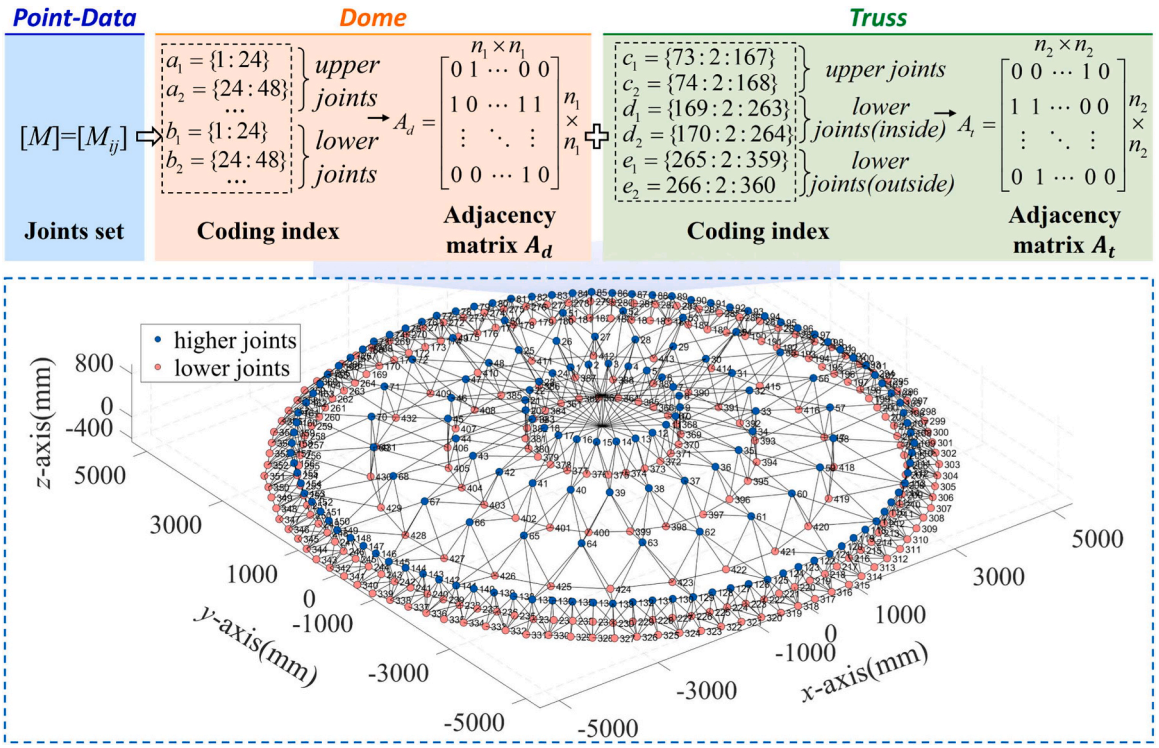


Fig. 17. Case: construction process of geometric digital twin of cable dome.

$$R(M) = R(A_d) + R(A_t) \tag{23}$$

5. Discussion

This section provides an in-depth discussion of the results, focusing on the effectiveness of the "three-point normal" cylindrical fitting model (T-Ncf), the accuracy of target recognition and joint fitting, the evaluation of joint deviation derived from gDT, the influence of module segmentation on joints accuracy, and the limitations of DS-gDT.

5.1. Performance analysis of the T-Ncf

Utilizing a scanned point cloud derived from a cylindrical steel pipe segment measuring 280 mm in length and 24 mm in radius as a case study (without any noise points), illustrated in Fig. 18(a), this paper introduces noise by incorporating a 0.24 ratio of noise points to a subset of the original point cloud, as depicted in Fig. 18(b), to emulate the complexities of real-world scanning environments. Subsequently, the RANSAC-based cylindrical surface fitting method and the proposed T-Ncf are applied to process the noise point cloud dataset shown in Fig. 18 (b). The dataset sizes vary across 6800, 14,630, 45,200, 84,500, and

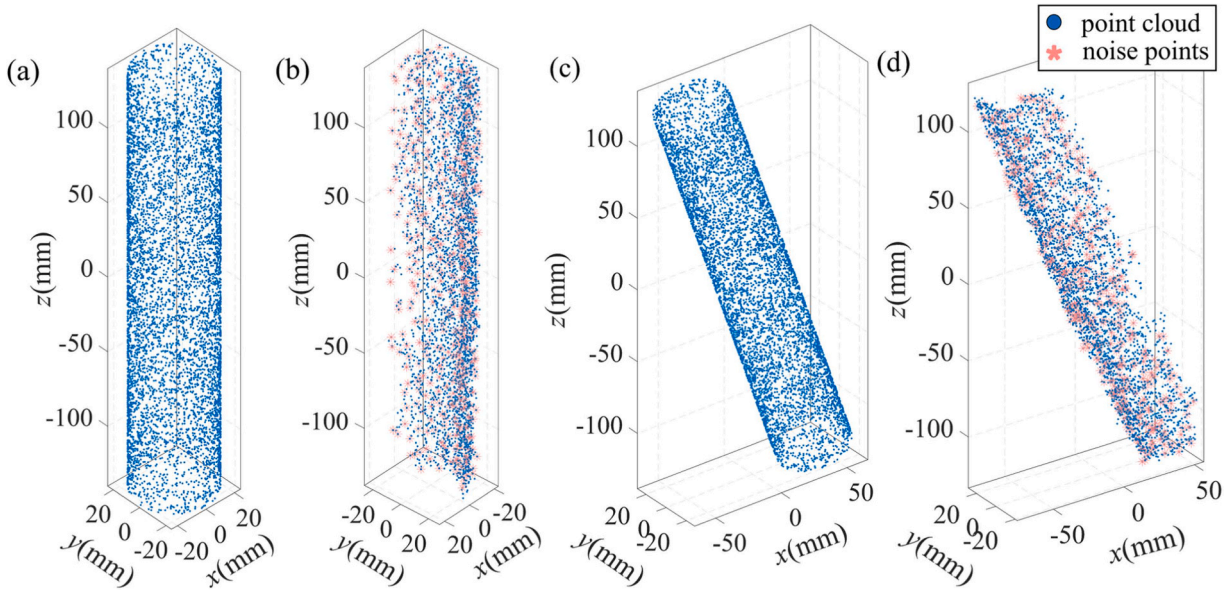


Fig. 18. Point cloud of tube: (a) Noise-free point cloud, (b) Simulated point cloud with noise, (c) Non-vertical noise-free point cloud, (d) Non-vertical point cloud with noise.

169,000 points. A comparative analysis is conducted on the processing time and the accuracy of radius fitting achieved by both methodologies across different dataset sizes. Additionally, random rotation angles were applied to orient the point cloud to the configuration shown in Fig. 18(c, d), aiming to validate whether different component orientations affect T-Ncf.

In Fig. 19(a), it is evident that the processing time of T-Ncf is consistently lower than that of the conventional RANSAC method across varying numbers of point clouds. Moreover, as the number of point clouds increases, the disparity in processing efficiency between the two methods progressively widens. As shown in Fig. 19(b), under the influence of noise points, both methodologies exhibit errors in radius fitting; however, the traditional RANSAC fitting demonstrates limited robustness with a maximum error of 18.8 %. In contrast, despite the impact of noise points on the results, the T-Ncf demonstrates superior stability and accuracy, with a maximum error of only 7.1 %. The calculation of T-Ncf on the point cloud in Fig. 18(d) similarly yields the above results, indicating that T-Ncf is not influenced by the orientation of the component.

5.2. Accuracy of extracting joints from PCD

This section conducts a comparative analysis of target segmentation and the precision of joint extraction. Initially, we contrast the target recognition and segmentation results between DS-gDT and the conventional RANSAC algorithm. The segmentation results of the traditional RANSAC algorithm applied to the point cloud of Module 1 are depicted in Fig. 14(b). Concurrently, a comparative analysis of the recognition and segmentation time for both methods is presented in Table 1. The findings reveal that the recognition rate achieved by the traditional RANSAC algorithm stands at approximately 65 %. Moreover, it is observed that the traditional RANSAC algorithm exhibits sensitivity to point cloud drift induced by interference occlusion and laser diffusion. In contrast, the component recognition success rate of DS-gDT surpasses that of the traditional RANSAC algorithm by approximately 51.5 %, with a corresponding efficiency improvement of about 20.09 %. Consequently, DS-gDT demonstrates superior performance in identifying dome structural components, displaying enhanced robustness and efficiency.

While DS-gDT is capable of fitting and extracting the spatial coordinates of all joints, the accuracy of its joint coordinate extraction necessitates further validation. Consequently, following the completion of tensioning the test model, the absolute coordinates of the joints are manually measured using methods such as a total station and general software fitting. To facilitate comparative analysis, the upper chord joint plane of the ring truss is set as the $z = 0$ plane, and the relative coordinates of all joints in relation to this plane are computed. Simultaneously, the joint coordinates extracted by DS-gDT undergo the same processing. Subsequently, the accuracy of DS-gDT coordinate extraction is evaluated using Eq. (24), where P_{re} denotes the coordinate extraction error, d_{pcd} represents the relative coordinates of joints extracted by DS-gDT, and d_{mea} represents the relative coordinates of joints measured manually.

Table 1
Execution time of point cloud recognition and segmentation.

Hardware capacity	Intel(R) Core (TM) i7-9700 CPU @ 3.00 GHz, 8 cores, 8 threads, 32.0 GB RAM
Project	Time (s)
RANSAC	64.812
DS-gDT	49.779

$$P_{re} = d_{pcd} - d_{mea} \tag{24}$$

A comprehensive statistical analysis was conducted to assess the error in coordinate extraction, with the findings visually represented in Fig. 20. The analysis revealed that the maximum coordinate extraction error occurs at the cable bracing joint, measuring 2.2 mm. Additionally, the coordinate extraction error at the lower chord joint within the ring truss exceeds 2 mm, amounting to 2.1 mm. Despite some joints exhibiting significant errors, it was observed that only 0.93 % of joint coordinates had errors exceeding ±2 mm, while 17.59 % had errors surpassing ±1 mm. Taking into consideration potential instrument inaccuracies and human-induced errors during measurements, a ±2 mm margin was established as the acceptable error range. Consequently, the accuracy of joint coordinates extracted by DS-gDT was determined to be above 99 %. This outcome not only validates the reliability of the proposed algorithm but also underscores its robustness in practical implementations.

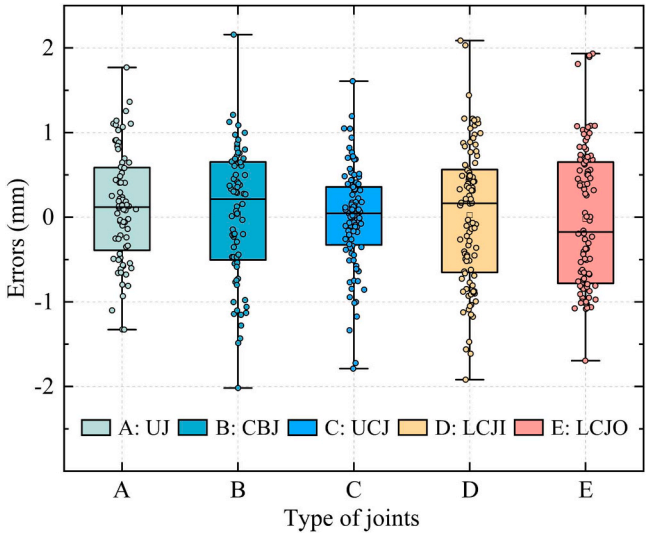


Fig. 20. Coordinate error between DS-gDT and manually measured.

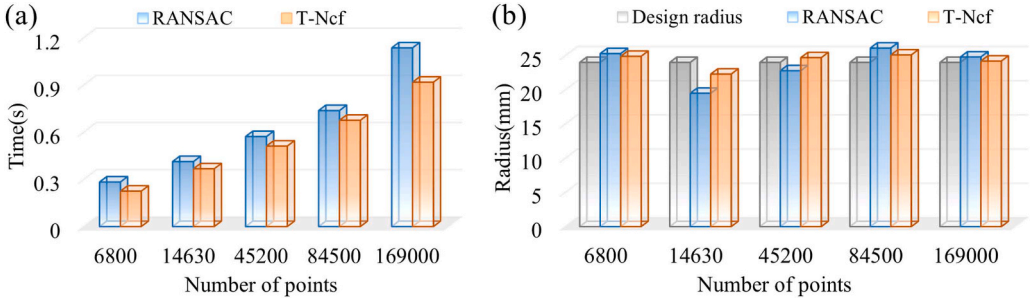


Fig. 19. Comparison of required time and fitting radius.

5.3. gDT-based joints deviation analysis

The gDT enables precise representation of the actual joint position post-structural tensioning. Through comparisons with the designed joint positions, efficient and accurate calculation of joint deviations is facilitated, offering a comprehensive and effective approach for joint deviation detection. This paper utilizes Eq. (25) to compute joint deviations, with the resulting calculations subjected to statistical analysis, as depicted in Fig. 21(a).

$$d_e = d_{pcd} - d_{des} \tag{25}$$

Here, d_e signifies the joints' deviations, with d_{pcd} denotes the relative coordinates of joints extracted by DS-gDT, and d_{des} represents the relative coordinates of design joints.

The analysis presented in Fig. 21(a) reveals significant deviations in the joints of the superstructure, with particular emphasis on the notable deviation observed at the second-lap upper joint, peaking at 33.11 mm. Furthermore, the maximum deviation at the cable bracing joint is also prominent in the second lap, measuring 31.27 mm. Specifically, the percentage of upper joint deviations exceeding 30 mm stands at 9.72 %, while those surpassing 20 mm amount to 33.33 %. Similarly, the proportion of cable bracing joint deviations exceeding 30 mm is 6.94 %, with deviations exceeding 20 mm accounting for 20.83 % of the total. Thus, the likelihood of significant deviations occurring in the upper joint is evidently greater. In the lower ring truss structure, the deviation of the upper chord joint is distributed within ± 10 mm, with 15.63 % of deviations exceeding 5 mm, and an average deviation of −0.61 mm. The maximum deviation of the inner joint of the lower chord reaches 12.67 mm, with 9.37 % of deviations exceeding 10 mm and an average deviation of 3.87 mm. Additionally, the maximum deviation of the outer joint of the lower chord is 14.862 mm, with 23.96 % of deviations exceeding 10 mm and an average deviation of 7.10 mm. This observation highlights the relatively pronounced deviation of the outer joint of the lower chord, attributed to various factors such as blanking length, welding temperature, and residual deformation.

Taking into account stability and other relevant factors, the permissible joint construction deviation for large-span cable dome structures is typically set at $L/300$ (where L represents the structural span) [32]. In this specific scenario, the limit value equates to 33.33 mm (i.e., 10,000/300). As the supporting structure for the upper dome assembly, the maximum permissible deviation for the ring truss structure is set at 30 mm [33]. Thus, despite notable construction deviations in the joints of both upper and lower structures, they still adhere to the specified deviation limits.

Furthermore, careful consideration is required for the radial deviation of the joints. Defining the uppermost joint as the origin $O(0, 0, 0)$, vector $\overrightarrow{OJ_s}$ is established using the design coordinates J (x_1, y_1, z_1) relative to O for each joint. Similarly, coordinates $\overrightarrow{OJ_m}$ are defined relative to measured joint coordinates and O. Subsequently, the radial deviation R_d of the joint can be computed using Eq. (26).

$$R_d = \left\|\overrightarrow{OJ_s}\right\| - \left|\frac{\overrightarrow{OJ_m}\cdot\overrightarrow{OJ_s}}{\left\|\overrightarrow{OJ_s}\right\|}\right| \tag{26}$$

Here, $\left\|\overrightarrow{OJ_s}\right\|$ represents the magnitude or length of vector $\overrightarrow{OJ_s}$.

Fig. 21(b) depicts the statistical distribution of radial deviations among joints. The maximum radial deviation for the upper joint occurs during the second lap, measuring 16.97 mm. Similarly, the maximum radial deviation for the cable bracing joint also occurs in the second lap, measuring 16.26 mm. Moreover, the percentage of upper joint radial deviations exceeding 10 mm is 15.28 %, while for the cable bracing joint it is 16.67 %. Hence, compared to joint deviations, larger radial deviations are less frequent. Importantly, the consistent location and proportion of maximum radial deviations for both upper and lower joints underscore DS-gDT's efficacy in capturing radial deviation synergy across corresponding parts.

In summary, this highlights the inevitability of notable joint deviations, underscoring the critical importance of acquiring and analyzing these variations from a holistic standpoint. The proposed method for joint deviation measurement based on gDTs exhibits significant potential for use in activities such as structural construction quality evaluations and early safety warnings.

5.4. The influence of duplicate fitting of identical joints

This section will discuss the effectiveness of the proposed method under the specific scenario of duplicate fitting of identical joints. In the point cloud of the cable dome structure, modules M_1 and M_2 are segmented (see Fig. 22a), and share common joints N_1, N_2, and N_3. These joints are accurately located using DS-gDT, as illustrated in Fig. 22 (b). It is worth noting that the proposed shape estimation model allows for the estimation of a cylindrical shape with a minimum of three points. Therefore, even if the steel tube is split longitudinally into two, accurate identification and segmentation of components can still be achieved. Eq. (14) is utilized for filtering out the fitting coordinates belonging to the shared joints. The average of these results serves as the final joint coordinate (Eq. 27). The joint design coordinates in the same coordinate system as the point cloud are as follows: N_1 (35, −1250, 795), N_2 (110, −2500, −125), N_3 (155, −3710, 340). The maximum error due to duplicate fitting of identical joints is calculated to be 1.8 %. Thus, despite instances of duplicate fitting of identical joints, the method exhibits robustness and stability. Nevertheless, effective module segmentation remains crucial as it mitigates the computational costs associated with redundant fitting of joints.

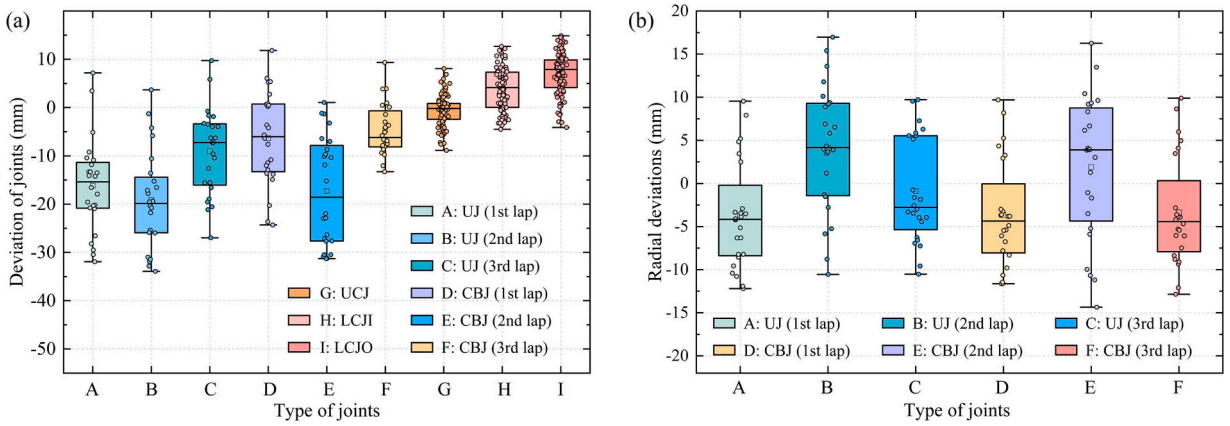


Fig. 21. (a) Deviations of joints. (b) Radial deviations of joints.

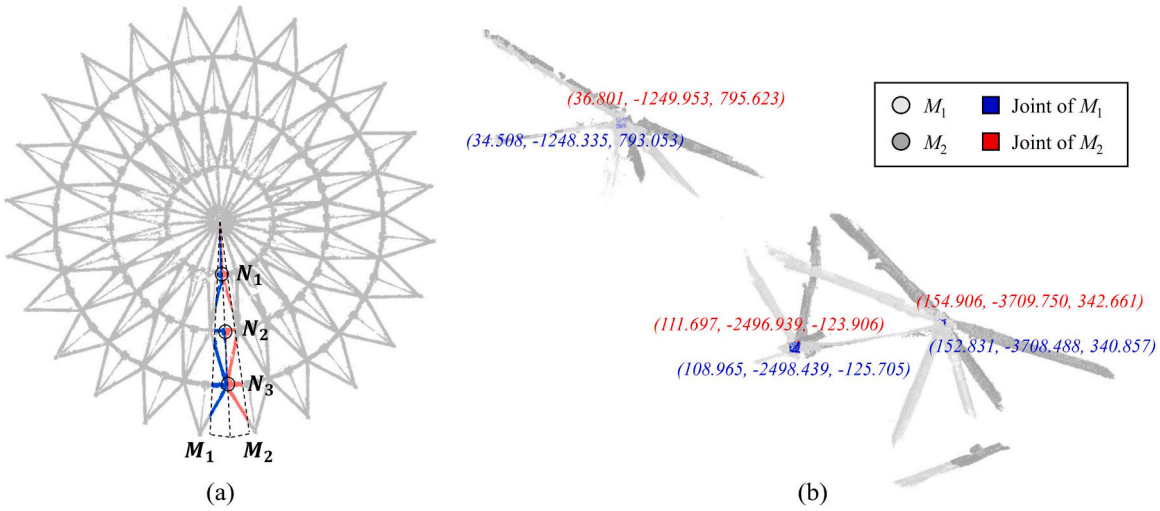


Fig. 22. Duplicate fitting of identical joints.

$$\begin{cases} \text{if} d \le d_{sj}, x = \sum x_i \big/ 2, y = \sum y_i \big/ 2, z = \sum z_i \big/ 2 \\ \text{elseif}, \varnothing \end{cases} \tag{27}$$

Where, d represents the deviation of two fitting results for shared joints, (x_i, y_i, z_i) represents the joint coordinates, $i = 1, 2$. d_{sj} represents the deviation limit, that is, the two points that meet this limit value are the duplicate fitting results of identical joints. This paper sets it at 20 mm.

5.5. Limitations of DS-gDT

Although DS-gDT proposed in this paper can effectively solve the problem of constructing the geometric Digital Twin of cable dome structures, this method has certain limitations. The limitations of this method are mainly reflected in the following aspects.

(1) According to the Standard for Design of Steel Structures (GB 50017–2017) [32], if the eccentricity between the brace member and the chord member connection joint of the intersecting joint (Fig. 23) does not exceed the limit specified in Eq. (28), the bending moment caused by eccentricity can be disregarded. Therefore, even when the intersection point of the component axes is non-singular, the joint position can still be determined using the method proposed in this paper. However, if the eccentricity exceeds the limit defined by Eq. (28), and when calculating the bearing capacity of the compressed chord member, the influence of eccentricity must be considered. In such cases, the eccentricity value is derived from design information. Automating the calculation of eccentricity remains a challenge that requires further resolution.

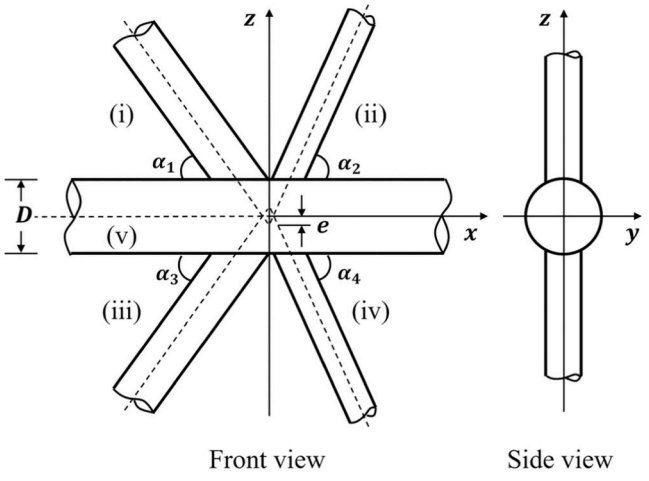


Fig. 23. The intersecting joint: (i)~(iv) is the brace member, (v) is the chord member.

$$-0.55 \le e/D \le 0.25 \tag{28}$$

Where, e denotes the eccentricity, D denotes outer diameter of the chord member.

(2) The proposed method in this paper addresses the issue of detecting joint coordinates in cable dome structures. However, in practical engineering applications, our method cannot monitor stresses/strains of structural components, nor can it update joint positions in real-time under varying loads. Processing massive point cloud inherently requires significant time.

(3) "Clustering" is primarily used for selecting "candidate points". However, during this process, potential bias may arise: in cases where there is a significant amount of missing point cloud data in paired regions, clustering can become challenging, thereby affecting the accuracy of subsequent joint fitting. Therefore, optimizing the laser scanning path is necessary to ensure comprehensive and high-quality point cloud data acquisition.

6. Conclusions and future work

In this paper, a method for establishing geometric digital twins (DS-gDT) for large-span cable dome structures is proposed, offering a comprehensive solution and technical pathway. The primary results are outlined as follows:

(1) DS-gDT demonstrates proficiency in automatically fitting joint coordinates while accurately identifying and segmenting pivotal components within PCD. The segmentation of components serves as the fundamental prerequisite for achieving precise joint localization. This segmentation process hinges on spatial positional relationships of components, with joints' positions determined by transitions between distinct components. In this study, the success rate of component segmentation is approximately 51.5 % higher compared to traditional methods. DS-gDT achieves joints fitting accuracy of 99 %. Furthermore, the automated fitting of joint coordinates, guided by component segmentation outcomes, plays a crucial role in the establishment of gDT, reducing the time investment required for gDT construction.

(2) DS-gDT represents an advancement and enhancement over conventional software and other gDT solutions available in the

market. The integration of the T-Ncf, sampling radius constraint mechanism, double termination criterion, and other methodologies significantly enhance the efficiency and precision of DS-gDT. Compared to traditional methods, DS-gDT achieves an efficiency improvement of approximately 20.09 %. Furthermore, by utilizing PCD as the sole input, DS-gDT effectively replaces the traditional labor-intensive measurement methods.

(3) This method offers a cohesive strategy for the visualization and digital management of gDTs. Leveraging a digital model automation algorithm rooted in the geometric topological relationships among joints, it enables seamless visualization of gDTs while facilitating digital representation and analysis of joint deviations. Moreover, this approach serves as a benchmark for early structural safety alerts based on joint deviations, showcasing extensive potential for widespread adoption and application.

In future research, the methodology presented in this paper will be extended to encompass a wider array of structural forms, thereby unlocking the potential for comprehensive geometric information management and analysis. This expansion may include applications to various structures such as suspend-dome structures, single-layer reticulated shell structures, and others. Building upon the gDT framework outlined in this paper, an advanced solution for digital twins of dome structure mechanics will be developed, further advancing the correlation between physical and mechanical property twins of structures. Additionally, leveraging the structural digital twin, there will be a quantitative exploration and analysis of the impact of joint deviations on structural stability, as well as the assessment of structural performance under diverse loading conditions.

CRediT authorship contribution statement

Guanghao Shangguan: Writing – review & editing, Resources. **Jie Wang:** Writing – original draft, Validation, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **yanxia zhang:** Writing – review & editing, Supervision. **Ailin Zhang:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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